

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

Problem-Solving and Data Analysis

17

9 answers · Rationale-rich · Teach from doc

Q1**C****C.** 320

Choice C is correct. It's given that the population density of Worthington is 290 people per square mile and Worthington has a population of 92,800 people. Therefore, the area of Worthington is $92,800 \text{ people} \times \frac{1 \text{ square mile}}{290 \text{ people}}$, which is equivalent to $\frac{92,800 \text{ square miles}}{290}$, or 320 square miles.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2**D****D.** Range of the high temperatures

Choice D is correct. The range of a data set is the difference between the maximum and the minimum values in the set. The maximum value among the high temperatures in the table is 82°F and the minimum value is 80°F. Therefore, the range is $82^\circ\text{F} - 80^\circ\text{F} = 2^\circ\text{F}$.

Choice A is incorrect. The median of a data set is the middle value when the values in the set are ordered from least to greatest. Ordering the high temperatures this way gives the list 80, 81, 81, 81, 82. Therefore, the median high temperature is 81°F. Choice B is incorrect. The mean high temperature is $\frac{81 + 80 + 81 + 81 + 82}{5} = \frac{405}{5} = 81$. Choice C is incorrect. The mode is the value that occurs the greatest number of times. For the set of high temperatures shown, 81 is the value that occurs 3 times, and therefore, 81°F is the mode of the high temperatures.

Q3**A****A.** 18

Choice A is correct. It's given that there are 2,358 raccoons in a 131-square-mile area. The estimated population density, in raccoons per square mile, is the estimated number of raccoons divided by the number of square miles. Therefore, the estimated population density of this area is $\frac{2,358 \text{ raccoons}}{131 \text{ square miles}}$, or 18 raccoons per square mile.

Choice B is incorrect. This is the number of square miles in the area, not the estimated number of raccoons per square mile in this area.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4**B****B.** 16

Choice B is correct. The mean of a set of data values is the sum of all the data values divided by the number of data values in the set. The sum of the data values shown is $2 + 9 + 14 + 23 + 32$, or 80. Since there are 5 data values in the set, the mean of the data shown is $\frac{80}{5}$, or 16.

Choice A is incorrect. This is the median, not the mean, of the data shown.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the maximum, not the mean, of the data shown.

Q5**39000**

The correct answer is 39,000. It's given that the giant armadillo has a mass of 39 kilograms. Since 1 kilogram is equal to 1,000 grams, 39 kilograms is equal to $39 \text{ kilograms} \times \frac{1,000 \text{ grams}}{1 \text{ kilogram}}$, or 39,000 grams. Therefore, the giant armadillo's mass, in grams, is 39,000.

Q6**14.1**

The correct answer is 14.1. It's given that a participant completes the bicycle race with an average speed of 24,816 yards per hour and 1 mile = 1,760 yards. It follows that this average speed is equivalent to $\frac{24,816 \text{ yards}}{1 \text{ hour}} \cdot \frac{1 \text{ mile}}{1,760 \text{ yards}}$, which yields $\frac{14.1 \text{ miles}}{1 \text{ hour}}$, or 14.1 miles per hour.

Q7**C****C.** 132

Choice C is correct. It's given that 3 teaspoons is equivalent to 1 tablespoon. Therefore, 44 tablespoons is equivalent to $44 \text{ tablespoons} \cdot \frac{3 \text{ teaspoons}}{1 \text{ tablespoon}}$, or 132 teaspoons.

Choice A is incorrect. This is equivalent to approximately 15.66 tablespoons, not 44 tablespoons.

Choice B is incorrect. This is equivalent to approximately 29.33 tablespoons, not 44 tablespoons.

Choice D is incorrect. This is equivalent to approximately 58.66 tablespoons, not 44 tablespoons.

Q8**A****A.** $13t$

Choice A is correct. It's given that the ratio x to y is equivalent to the ratio 12 to t . This can be represented by $\frac{x}{y} = \frac{12}{t}$. Substituting 156 for x in this equation yields $\frac{156}{y} = \frac{12}{t}$. This can be rewritten as $12y = 156t$. Dividing both sides of this equation by 12 yields $y = 13t$. Therefore, when $x = 156$, the value of y in terms of t is $13t$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9

B

B. 23

Choice B is correct. It's given that 100 centimeters is equal to 1 meter. Therefore, 2,300 centimeters is equivalent to $2,300 \text{ centimeters} \times \frac{1 \text{ meter}}{100 \text{ centimeters}}$, or 23 meters.

Choice A is incorrect. 0.043 meters is equivalent to 4.3, not 2,300, centimeters.

Choice C is incorrect. 2,400 meters is equivalent to 2,400, not 2,300, centimeters.

Choice D is incorrect. 230,000 meters is equivalent to 230,000, not 2,300, centimeters.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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